

Charles Xue

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EDUCATION

Cornell University, College of Arts & Sciences

Expected May 2028

B.A in Computer Science and Statistics

Ithaca, NY

GPA: 4.0 | **Relevant Coursework:** Functional Programming, Discrete Math, Data Structures & Algorithms

EXPERIENCE

Generative AI at Cornell

Feb. 2026 – Present

Project Engineer

Ithaca, NY

- Architected an end-to-end student outreach pipeline automating lead discovery and personalized email generation across 10k+ Apollo-sourced startup contacts, reducing per-lead research and outreach time by 95%.
- Integrated the Anthropic API to research startups and generate context-aware cold emails matching user style.

Cornell Cybersecurity Club

Sep. 2025 – Present

Lead Developer

Ithaca, NY

- Engineered the agent communication layer of a Go/C++ command-and-control framework for competition use, designing the implant-to-server handshake protocol via protocol buffers to manage concurrent sessions across 10+ simulated hosts.

PROJECTS

CVE Intel — [cve-intel](#) | [live](#)

Python, FastAPI, Next.js, TypeScript, PostgreSQL, pgvector, sentence-transformers, Anthropic API

- Built a RAG-powered platform for natural-language CVE vulnerability search, ingesting 64k NVD CVEs into PostgreSQL and pgvector with sentence embeddings indexed for sub-millisecond approximate nearest-neighbor retrieval.
- Designed a three-strategy query router (regex exact lookup → hybrid SQL + vector search → pure ANN fallback) that grounds responses via the Anthropic Citations API and strips any hallucinated CVE IDs absent from the retrieved context.

Philly Vibe Map — [philly-vibe-map](#) | [live](#)

Python, FastAPI, React, TypeScript, DistilBERT, sentence-transformers, FAISS, MapLibre GL

- Fine-tuned DistilBERT with LoRA for neighborhood sentiment classification, then ran a sentence-transformer + BERTopic + FAISS pipeline over 1.1M Yelp reviews to score 157 Philadelphia neighborhoods across 6 vibe dimensions.
- Deployed an interactive map visualizing year-over-year vibe shift across 157 Philadelphia neighborhoods from 2007 to 2022.

Podium — [podium](#) | [live](#)

TypeScript, React, MediaPipe, Whisper.wasm, IndexedDB

- Built an on-device presentation coach analyzing 6 behavioral dimensions in real-time powered by a Web Worker ML pipeline orchestrating 3 concurrent MediaPipe models for gaze detection, gesture tracking, and expressiveness scoring.
- Implemented a weighted scoring engine aggregating each dimension into a composite presentation score, with two-pass filler word detection (Web Speech API + Whisper.wasm) and per-user ML calibration.

Coursemap — [coursemap](#)

TypeScript, Next.js, React, PostgreSQL, Anthropic API

- Built a Canvas-to-Google Calendar sync app with a three-tier event classifier (regex → shared DB cache → Claude Haiku) and cross-user caching, cutting redundant AI calls to near-zero for recurring event patterns.
- Reduced GCal API calls to one per sync via HashMap deduplication on Canvas UID against a local DB snapshot.

SkyOps

TypeScript, Next.js, Anthropic API, Supabase, React

- Developed a charter flight operations platform automating flight quoting and pricing, with a fleet utilization forecasting engine using a multiplicative demand model with day-of-week seasonality and probability-weighted pipeline demand.
- Engineered a recursive multi-leg route optimizer with greedy fuel-stop selection and a Dijkstra fallback to minimize empty-leg flights and aircraft repositioning costs; validated against a mock dataset of 2,000 flights.

TECHNICAL SKILLS

Languages: Python, TypeScript, Go, Java, C++, JavaScript, SQL, Bash, R

Tools: Docker, PostgreSQL, Linux, Git, GitHub, Vercel, Supabase

Frameworks & Libraries: FastAPI, React, Next.js, Electron, Anthropic API

ML & Data: PyTorch, sentence-transformers, FAISS, BERTopic, LoRA, MediaPipe, pgvector